A Multi-domain, Artifact-aware, and Explanation-integrated Universal Synthetic Image Detection

Capstone Project Milestone Report

Author: Tanvir Mahmud Prince, Eshan Agarwal

Course: Advanced Machine Learning (ECE 5424)

Github Repo: https://github.com/ToonLink123/Capstone-Project-Adv-ML, https://github.com/Tanvir-13EEE/Capstone-Project-Adv-ML2.git

1. Introduction

This milestone report presents the progress toward developing a universal synthetic image detection framework capable of accurately distinguishing AI-generated images from real photographs while providing interpretable explanations. The project is progressing in alignment with the proposed timeline and has successfully reached the Week 5–6 development phase, demonstrating steady advancement toward the overall research objectives.

2. Progress From Proposal

According to the original project timeline, Weeks 1–2 were dedicated to dataset collection and preprocessing. Two datasets, the CiFake Dataset and FaceForensics++, have been utilized for initial analysis. The CiFake dataset contains approximately 60,000 images across diverse non-human categories (e.g., animals, vehicles, and natural scenes), while FaceForensics++ comprises around 7,000 images and videos focused on human facial manipulations. Together, these datasets provide a balanced combination of real and synthetic visual data, making them well-suited for evaluating synthetic image detection approaches. All images were preprocessed through resizing and normalization to ensure compatibility with deep learning models.

During Weeks 3–4, baseline models were implemented in accordance with the proposed methodology. These include:

(i) a spatial-domain classifier based on a conventional convolutional neural network,

(ii) a frequency-domain classifier utilizing Fast Fourier Transform (FFT)-transformed inputs to capture spectral artifacts,

(iii) a transfer learning approach leveraging the pretrained VGG16 model to distinguish between real and synthetic images, and

(iv) the integration of a lightweight large language model to generate interpretable explanations of detected artifacts.

Among these, the VGG16-based model has been fully implemented and trained using the CiFake dataset. The developed pipeline includes data preprocessing, model training, performance evaluation through confusion matrix analysis, visual interpretation using Grad-CAM, and an LLM-based module for artifact-level explanation.

Currently, the project is progressing toward the integration of spatial and frequency-domain features into a unified detection framework. Although the complete fusion architecture has not yet been finalized, all essential components required for its implementation have been successfully developed, establishing a strong foundation for the next phase of the project.

3. Dataset Analysis

The FaceForensics++ dataset is structured using CSV metadata files that specify file paths, labels, and video properties. For preprocessing, one representative frame is extracted from each video using OpenCV. Frames are resized and converted into tensors for input into neural networks.

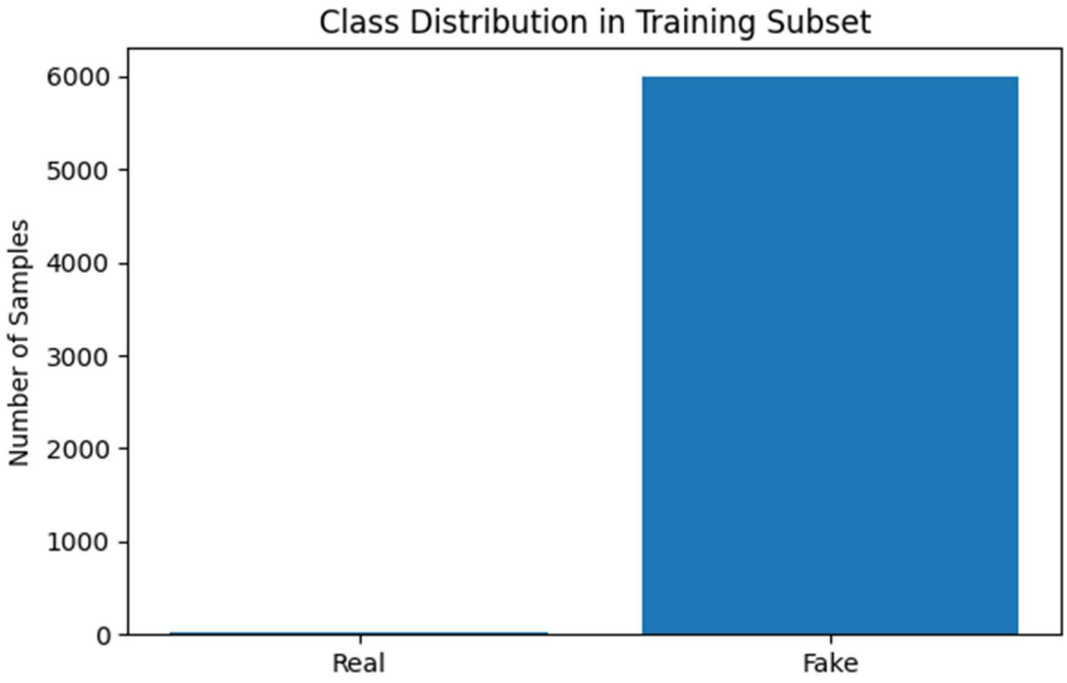


Figure 1. Distribution of Real and Synthetic Images in FaceForensics++ Dataset

First, the dataset composition was analyzed by counting real and fake samples across manipulation methods (DeepFakeDetection, Deepfakes, Face2Face, FaceShifter, FaceSwap, and NeuralTextures). A class distribution plot revealed that the dataset is skewed toward manipulated samples in the current subset, which may affect model behavior. Second, video metadata such as frame count, resolution, and file size were examined. Most videos have a resolution of 1920×1080, while frame counts and file sizes vary significantly across manipulation types. This suggests that single frame sampling may not fully capture temporal diversity. Third, qualitative analysis was performed using frequency domain representations. FFT magnitude plots of real and fake frames show differences in spectral patterns, supporting the hypothesis that frequency-domain features can help identify synthetic artifacts.

On the other hand, The CiFake dataset was used for training and evaluation of VGG16 model. The dataset contains real images and AI-generated images collected from multiple generative sources. Images were organized into training, validation, and testing sets. All images were resized to 224×224 pixels to match the VGG16 input requirements.

Figure 1. Sample Images in CiFAKE Dataset

Exploratory dataset inspection verified class balance and ensured that both real and synthetic samples were properly represented in each subset. Proper dataset organization is critical for ensuring unbiased model evaluation.

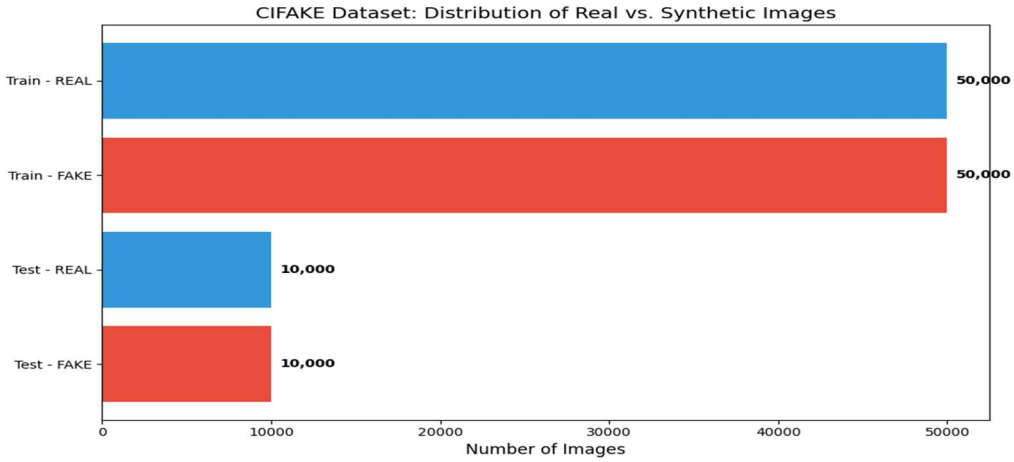


Figure 2. Distribution of Real and Synthetic Images in CiFAKE Dataset

4. Preliminary Results:

Two parallel processing pipelines were implemented:

1. Spatial Pipeline: RGB frames are directly used as input to a convolutional neural network (CNN) for classification.

2. Frequency Pipeline: Grayscale frames are transformed using a 2D Fast Fourier Transform (FFT), and the log magnitude of the frequency spectrum is used as input to a CNN. This representation shows high-frequency artifacts that are often present in manipulated media.

Both classifiers use the same CNN architecture to ensure a controlled comparison between spatial and frequency representations. The training pipeline includes dataset splitting, batch loading, and evaluation using accuracy, precision, recall, and F1-score.

The codebase is organized into modular components including dataset loading, model definition, FFT transformation, training, and evaluation. This structure will allow for future extensions such as multi-frame processing and model fusion.

Table: Model Performance Across Experimental Configurations

Model	Epochs	Max Fake/CSV	Accuracy	Precision	Recall	F1 Score
Spatial CNN	3	8	88.24%	88.24%	100.00%	93.75%
Frequency FFT-CNN	3	8	88.24%	88.24%	100.00%	93.75%
Spatial CNN	5	8	76.47%	76.47%	100.00%	86.67%
Frequency FFT-CNN	5	8	76.47%	76.47%	100.00%	86.67%
Spatial CNN	3	3	66.67%	62.50%	100.00%	76.92%
Frequency FFT-CNN	3	3	44.44%	44.44%	100.00%	61.54%

Overall, these experiments demonstrate that while both spatial and frequency models can achieve high performance under imbalanced conditions, their generalization ability is limited. The results highlight the strong influence of dataset composition and training configuration on model behavior.

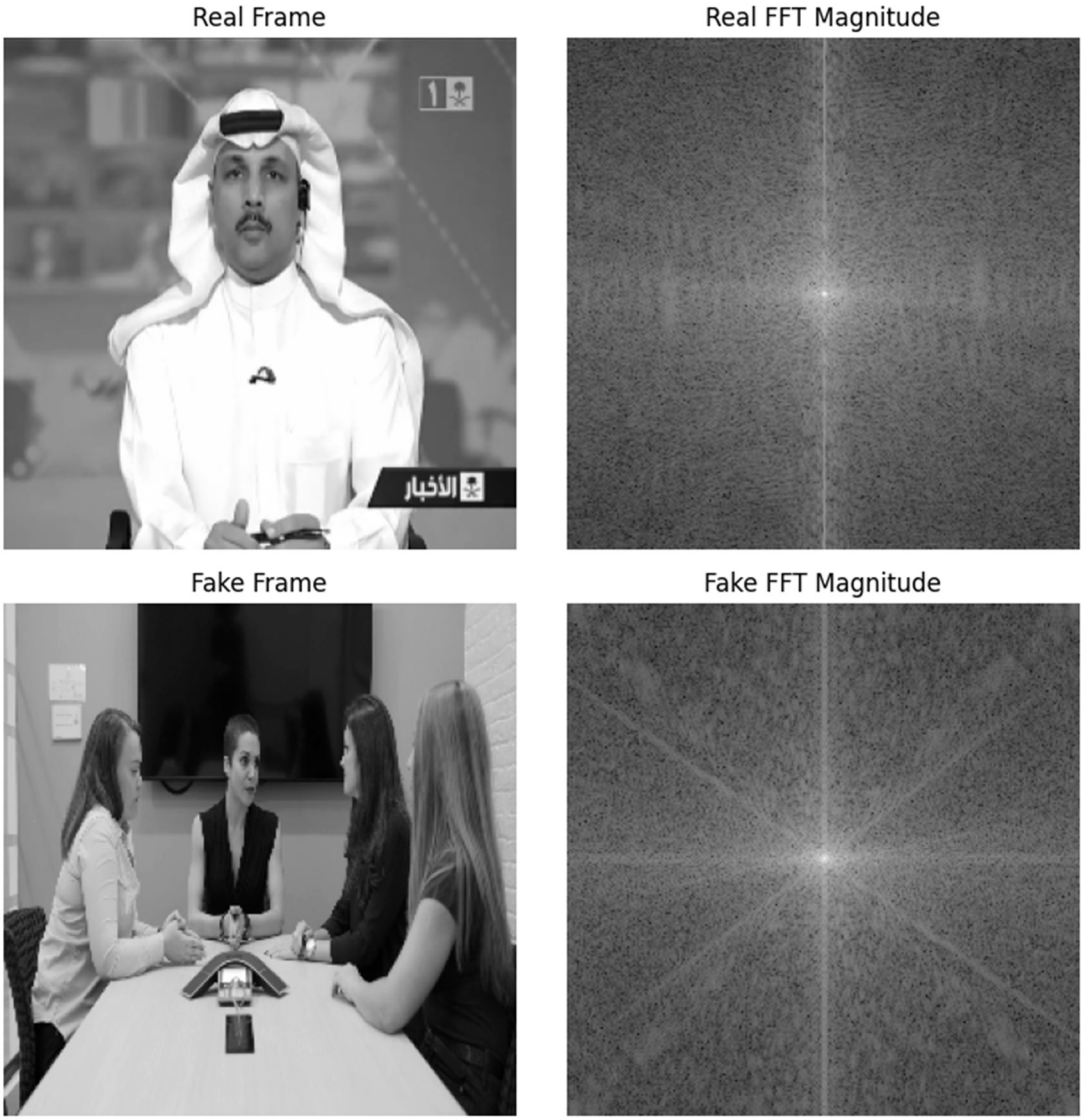
Confusion matrix:

[[0, 4],

[0, 13]]

The confusion matrix shows that all validation samples were classified as fake. While all fake samples were correctly identified (true positives), all real samples were misclassified as fake (false positives). This results in high recall but poor specificity.

A challenge identified during EDA is class imbalance and limited diversity in real samples. This is reflected in the confusion matrix results, where both models predicted all validation samples as fake. Additionally, the use of a single frame per video may limit the ability of the model to generalize.



5. Study on Transfer learning

A transfer learning strategy was adopted using the pretrained VGG16 convolutional neural network. The convolutional backbone was initialized with ImageNet weights and used as a feature extractor. Custom fully connected layers were added for binary classification of real versus AI-generated images.

Table 2. Training Hyperparameters

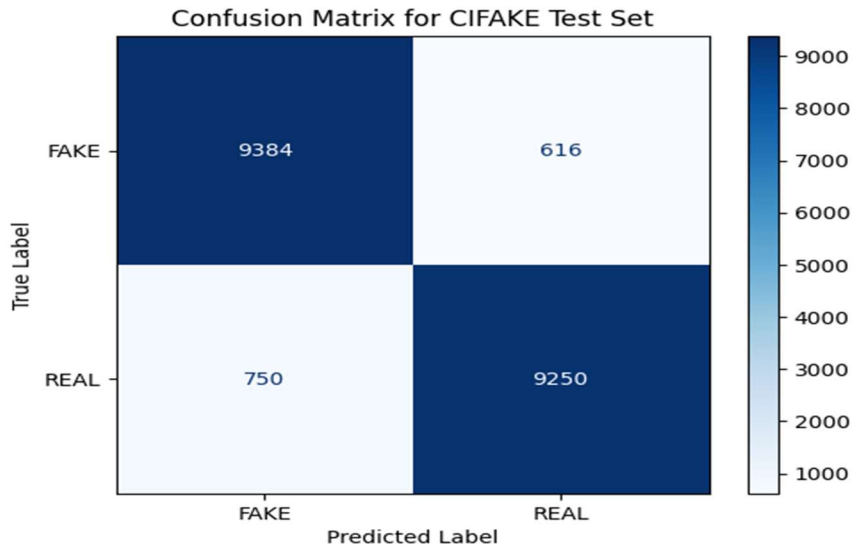
Parameter	Value
Model Backbone	VGG16 (Pretrained ImageNet)
Loss Function	Binary Cross Entropy
Optimizer	Adam
Task	Binary Classification (Real vs Fake)
Total params	14,846,273
Trainable params	131,585

a. Model Training

The model was trained using supervised learning. During training, the network learns spatial texture patterns that differentiate authentic photographs from synthetic images generated by AI systems. Training and validation metrics were monitored across epochs to evaluate convergence.

b. Model Evaluation

Model performance was evaluated using a held-out test dataset. Evaluation metrics included classification accuracy and confusion matrix analysis. These metrics provide insight into the model's ability to correctly classify real and fake images. Confusion matrix analysis revealed that most samples were correctly classified, indicating that the model successfully captured distinguishing visual patterns between real and synthetic images. False positives occur when real images contain unusual textures that resemble synthetic artifacts, while false negatives occur when generated images appear highly realistic with minimal visible artifacts. VGG-16 model achieves 93% accuracy to predict real vs fake image for CiFAKE Dataset.



c. Explainability Using Grad-CAM

To interpret model decisions, Grad-CAM visualization was applied to highlight image regions that contributed most strongly to the classification output. The method computes gradients with respect to convolutional feature maps and produces a heatmap indicating influential spatial regions.

Experimental observations indicate that the VGG16 model primarily relies on spatial texture features when distinguishing synthetic images. Grad-CAM visualizations suggest that the network focuses on regions containing subtle inconsistencies such as irregular high-frequency textures, unnatural edges, or repeated structural patterns.

These patterns are consistent with known artifacts introduced by generative models. However, analysis also shows that some synthetic images contain fewer detectable artifacts, making them more difficult to classify correctly. This observation supports the need for additional frequency-domain analysis and cross-domain feature fusion.

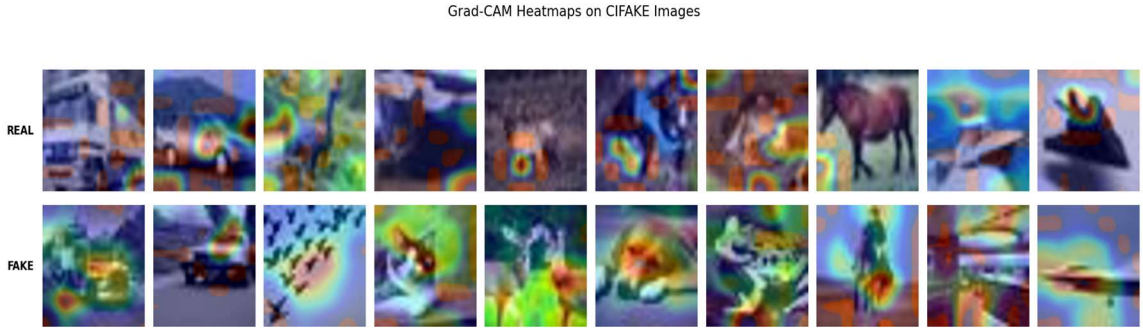


Figure 6. Visualization illustrating artifact localization within synthetic imagery.

6. Future Work

The next phase of this project will focus on improving model generalization and addressing the limitations identified during the preliminary experiments. Several targeted enhancements are planned to strengthen the overall detection framework:

- **Class Imbalance Mitigation:**
  To reduce bias toward the dominant class (manipulated samples), techniques such as weighted loss functions, balanced sampling strategies, and data augmentation will be employed to ensure a more representative training distribution.
- **Hybrid Spatial–Frequency Modeling:**
  A unified model will be developed that integrates both spatial-domain features and frequency-domain representations using Fast Fourier Transform (FFT). This hybrid approach aims to improve robustness by capturing complementary artifact patterns.
- **Dataset Expansion:**
  The dataset will be further enriched by incorporating additional samples from FaceForensics++, thereby increasing diversity and improving the model's ability to generalize across different manipulation techniques.
- **Hyperparameter Optimization:**
  Systematic tuning of key hyperparameters—such as learning rate, batch size, and training duration—will be conducted to enhance performance while minimizing overfitting.